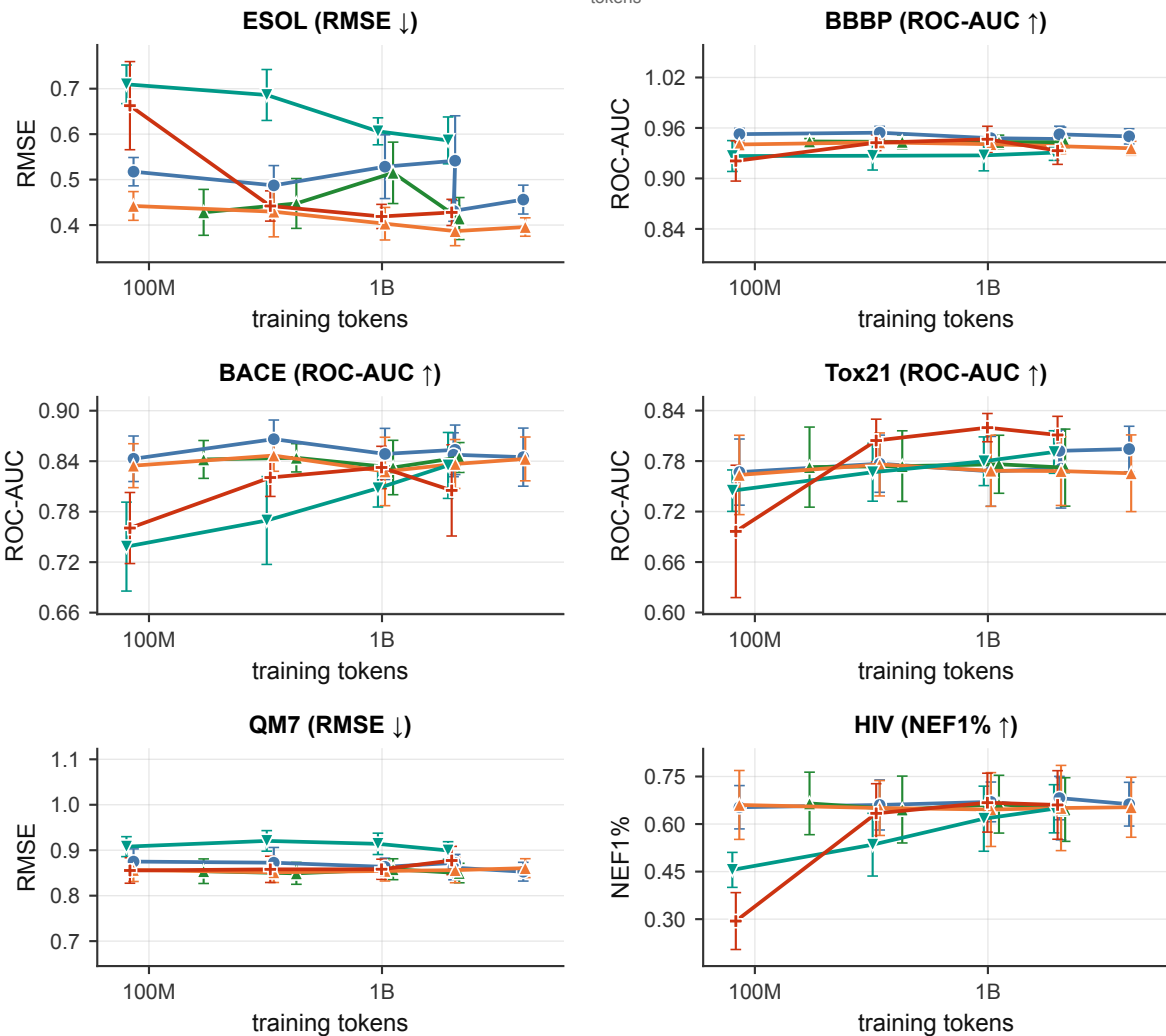


Fig A2.a - scaling of the primary regimes in training tokens

pooled 5-fold scaffold CV · error bars = ± 1 sd across the 5 folds (no head-seed, no pretraining-seed spread), so every interval means the same thing · the classical anchors and no_pretrain baselines are omitted for legibility — they are bars in A1.a/A1.b · x = tokens actually processed (trainer's own non-padding count from metrics.jsonl, per run — NOT forward passes $\times$ a constant; that ratio varies 40.0–42.9 tok/seq across corpora) · unsup $\rightarrow$ sup = MLM-base tokens + SFT tokens



● unsup_only (MLM) ▲ sup_only: dense (MTR) ■ sup_only: dense+sparse
 ▲ unsup $\rightarrow$ sup: dense (MTR) ▼ sup_only: sparse_all